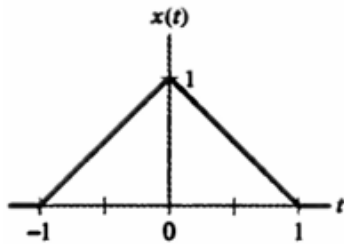


Question. 1 (CLO 1) (10 Marks)

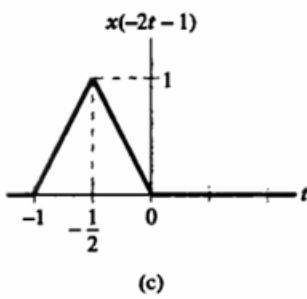
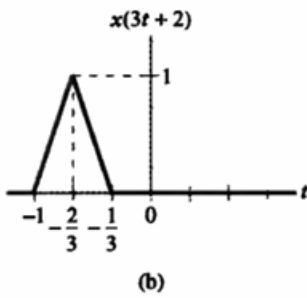
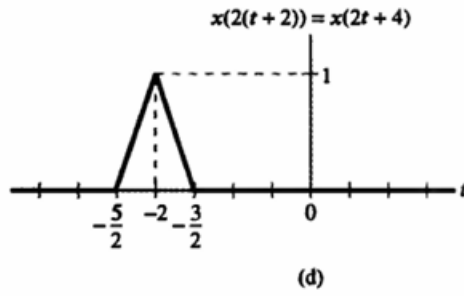
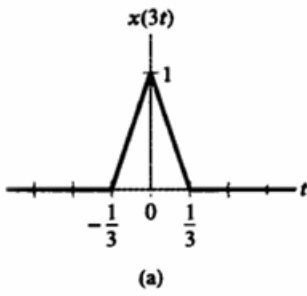
Q1: Express the types of operations involved in the given signal $x(t)$ for the following transformations. Also, sketch the resulting signals derived from $x(t)$. [2.5+2.5+2.5+2.5 marks]



- (a) $x(3t)$
- (b) $x(3t+2)$
- (c) $x(-2t-1)$
- (d) $x(2(t+2))$

Solution

Signals	Operations
$x(3t)$	Time Compression by a factor 3
$x(3t+2)$	Time Compression & Time Shifting
$x(-2t-1)$	Time Compression, Time Reversal & Time Shifting
$x(2(t+2))$	Time Compression & Time shifting



Question #2

Q-11 ∞ Express the convolutional process mathematically

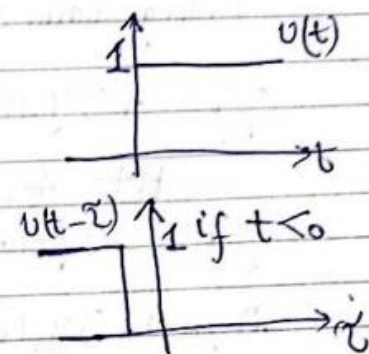
$$y(t) = e^{\lambda t} u(t) * u(t) = \int_0^t e^{\lambda \tau} u(\tau) u(t-\tau) d\tau$$

As $u(t-\tau) = u(t) = 1$

Therefore

$$y(t) = e^{\lambda t} u(t) * u(t) = \int_0^t e^{\lambda \tau} (1) (1) d\tau$$

$$= \int_0^t e^{\lambda \tau} d\tau$$



By integrating

$$y(t) = e^{\lambda t} u(t) * u(t) = \frac{1}{\lambda} \left[e^{\lambda \tau} \right]_0^t$$

$$= \frac{1}{\lambda} \left[e^{\lambda(t)} - e^{\lambda(0)} \right]$$

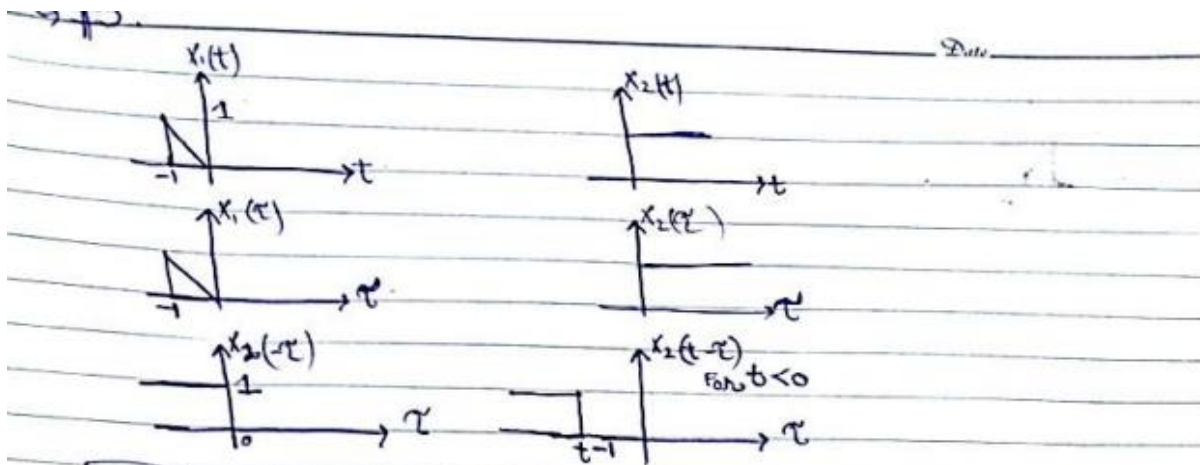
$$= \frac{1}{\lambda} [e^{\lambda t} - 1]$$

$$y(t) = e^{\lambda t} u(t) * u(t) = -\frac{1}{\lambda} [1 - e^{\lambda t}]$$

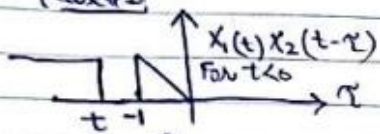
We can also write as

$$y(t) = e^{\lambda t} u(t) * u(t) = -\frac{1}{\lambda} [1 - e^{\lambda t}] u(t)$$

Question # 3

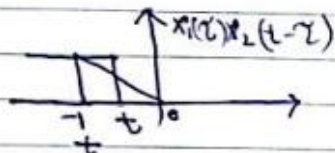


Case #1



$$y(t) = \int_{-\infty}^{\infty} x_1(\tau)x_2(t-\tau) d\tau \Rightarrow 0 \text{ As no overlap.}$$

Case #2

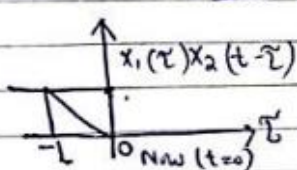


$$y = \int_{-1}^t -\tau d\tau \text{ As } u(\tau) = 1.$$

$$y = \left[-\frac{\tau^2}{2} \right]_{-1}^t = \left[-\frac{\tau^2}{2} \right]_{-1}^t$$

$$y = -\frac{t^2}{2} + \frac{(-1)^2}{2} \Rightarrow \frac{1}{2} - \frac{t^2}{2}$$

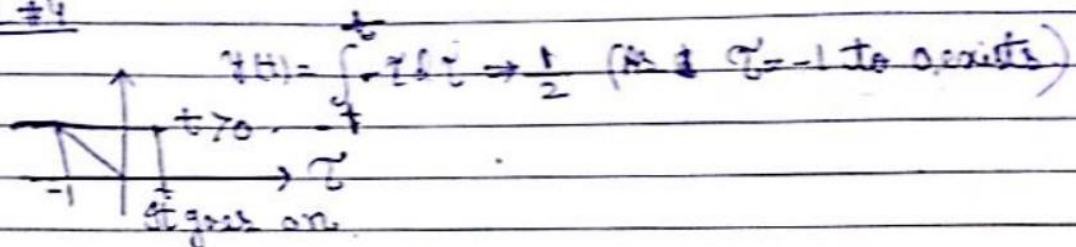
Case #3



$$y(t) = \int_{-1}^0 -\tau d\tau$$

$$y(t) = \left[-\frac{\tau^2}{2} \right]_{-1}^0 \Rightarrow \frac{1}{2}$$

Case #1



$$y(t) = \frac{1}{2}$$

$$y(t) = \begin{cases} 0 & t < -1 \\ \frac{1}{2} - \frac{t^2}{2} & -1 \leq t \leq 0 \\ \frac{1}{2} & t > 0 \end{cases}$$

